

Design of True Random Number Generator Based on Multi-Ring Convergence Oscillator Using Short Pulse Enhanced Randomness

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Abstract—The entropy source structure with embedded XOR gates in a ring oscillator (RO) as a true random number generator (TRNG) can improve the speed of accumulating jitter in the oscillator. However, the XOR gate has a certain response time to the input change, and when the input changes too fast, the XOR gate will output short pulses. In this paper, we propose a TRNG design based on a multi-ring convergence oscillator (MRCO) making use of the characteristics of short pulses. We study the output of the XOR gate when facing different inputs. By modeling the time of a fibonacci ring oscillator (FIRO) as an example, we find that the loss of short pulses in an inverter chain is the reason for making the FIRO enter into periodic oscillation. This phenomenon suppresses the accumulation of jitter and occurs periodically in existing structures. Our proposed structure uses independent sub-rings to accumulate jitter, allowing the main-ring to quickly generate short pulses to provide analog randomness. The proposed TRNG design is implemented in Xilinx Virtex-6 FPGA. The experimental results show that it has the highest ratio of throughput rate to hardware resources. The generated random sequence pass both NIST SP800-22 test and NIST SP800-90B test.

Index Terms—True random number generator, XORs, FIRO, short pulse, multi-ring convergence oscillator.

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I. INTRODUCTION

IN THE information age, personal and private information are transmitted or stored with smart devices, and the security of such information is extremely important. Cryptographic keys [1], initialization vectors, hashes and nonce all of which rely on random numbers, play an critical role in securing information. Hence, a reliable source of randomness must be provided in security systems. Random number generators are classified as pseudo random number generators (PRNGs) and true random number generators (TRNGs). The PRNG creates random numbers with a finite period through algorithms and seeds, which are vulnerable to malicious attacks [2]. On the other hand, TRNG extracts independent random numbers from random physical processes; as a result, no one can predict the sequence of random numbers generated by the TRNG [3].

Circuit structures that use physical processes to provide a source of randomness for TRNGs are called entropy sources. Common physical processes include metastable state [4], [5], [6], [7], clock jitter [8], [9], [10], [11], chaos [12], and quantum phenomena. The entropy sources are usually sampled using D flip-flops, and sampled raw random numbers often require additional post-processing circuitry to improve their statistical properties in order to pass the randomness test. To build a high-quality TRNG, most of the existing methods require significant hardware resources, and deploying TRNGs in resource-constrained portable devices poses a severe challenge. Field Programmable Gate Arrays (FPGAs) are an important part of embedded digital systems, and building TRNGs in FPGAs is a very efficient way to design for fast TRNG deployments [13]. Two physical processes exist in FPGAs that are easily accessible and have randomness. One is the metastable state and the other is the clock jitter.

Metastability can be generated by making a flip-flop setup or hold violation or by using SR latches with the same logic level input at the same time. The output of these circuits may be ‘0’ or ‘1’ when metastability occurs, and the randomness comes from the noise inside the circuit. However, a TRNG based on metastability needs to be carefully designed to meet the requirements of symmetric layout; otherwise, the output random number will have a large bias. Danger et al. proposed to replace a flip-flop with a latch built from a look-up

table (LUT) since flip-flops are usually built to minimize metastability during fabrication [5]. Delay line can also be used to make a latch meet the condition of entering the metastable state. This method is very demanding in terms of wiring. Li et al. suggested to use cross-coupled NAND gate gates and manual wiring to generate meta stable events [6]. The output throughput rate can reach 30 Mbps; however, additional calibration feedback circuits and von Neumann post-processing circuits are required to improve the randomness of the generated numbers. The disadvantage of using metastability is that it requires difficult manual wiring and post-processing circuitry to reduce metastability output bias cause by process bias.

Clock jitter is a slight shift in the transition time of logic gate potentials affected by noise. Sunar et al. suggested to use multiple ring oscillators (ROs) for heterodyning in order to ensure that each output bit can contain clock jitter and through post-processing circuitry to output a high entropy random sequence [14]. However, this again requires a large amount of hardware resources. To mitigate the drawbacks of the Sunar structure, Wold et al. proposed to sample the output of a RO before passing it through an Exclusive-OR (XOR) tree. This can improve the XOR tree that has difficulty in handling a large number of transitions from a RO [15]. Golic proposed to replace ROs as the entropy source with more complex Fibonacci ring oscillators (FIROs) with feedback and Galois ring oscillators (GAROs). It was demonstrated that embedding an XOR gate in a RO can effectively increase the oscillation speed of the RO, accelerate the accumulation rate of jitter, and generate a high-speed oscillation signal with randomness in digital and analog levels. As a result, the flip-flops used for sampling can also enter the metastable state enabling the generation of the high-speed raw binary sequence with higher entropy rate. This approach overcomes the coupling problem in ROs and increases the source of randomness in a TRNG [16]. However, Dichtl's study showed that using an FIRO as an entropy source structure has the risk of entering into periodic oscillations. When this occurs, the entropy of an FIRO will be significantly reduced [17]. Recently, Yang et al. proposed a novel FIRO/GARO based chaotic oscillation ring structure to form reliable chaotic oscillations [18]. Mei et al. argued that XOR gates can shorten the time of jitter accumulation and proposed chained oscillatory rings using XOR gates nested with multiple ROs [19]. Works of Yang and Mei effectively improved the performance of the entropy source structure and demonstrated the great potential of XOR gates in improving the randomness of entropy sources. However, they did not take into account the phenomenon that the XOR gate outputs short pulses that do not reach the full level when fed with a high-speed changing input signal. Such short pulses may be eliminated during the propagation within the oscillator, lowering the accumulated rate of oscillator jitter and reducing the generation possibility of metastability. Therefore, it is highly necessary to study the relationship between the short pulses from an XOR gate and its input signals as well as the output behavior of the XOR gate in an oscillator in order to utilize the XOR gate more efficiently

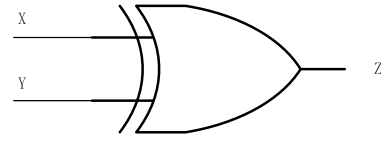


Fig. 1. Logic symbols of an XOR.

to improve the oscillator-based TRNG randomness to improve its security.

In this paper, a lightweight and high-throughput TRNG design is proposed. The proposed TRNG structure is verified on Xilinx Virtex-6 and Spartan-6 FPGAs with sampling rates of 350 MHz and 500 MHz for the entropy source structure, respectively. The collected data are evaluated by the National Institute of Standards and Technology (NIST) SP800-22 and SP800-90B [21], [22]. The proposed TRNG is evaluated against various attacks, and the experimental results show that the proposed TRNG is resistant to external temperature and voltage variations and insensitive to frequency injection.

The main contributions are summarized as follows:

- 1) The jitter accumulation phenomenon based on XOR gate models is analyzed, and it reveals that XOR gate will generate short pulses, which helps obtaining more random entropy.
- 2) Based on the XOR gate jitter model analysis, this paper build FIRO time model of pulse propagation, which can reduce the random phenomenon that short pulse elimination in the FIRO. By analyzing the mechanism that brings FIRO into periodic oscillation, we also describe several design principles for enhancing the FIRO performance as a TRNG.
- 3) A TRNG based on multi-ring convergence oscillators (MRCOs) is proposed, which enables faster generation of short pulses and allows more randomness to be accumulated in the time difference of XOR gate inputs for generating short pulses. The main-ring is completely composed of two-input XORs, with one input of each XOR gate connected to other XOR gates and the other input connected to the sub-ring. The sub-ring is a ring oscillator consisting of inverters.

The rest of this paper is organized as follows. In Section II, we present a study on the behavior of XOR gates and FIRO oscillations. Section III presents the proposed TRNG structure and its randomness analysis. Section IV presents the implementation details of the proposed structure on FPGAs and evaluation. Section V summarizes this paper.

II. PRELIMINARY STUDY

A. Behavior Model of XOR

The XOR gate in Fig. 1 is a common logic gate in combinational logic circuits. It is widely used in the design process of true random generators by combining multiple random bit outputs to improve the randomness of the entropy source output and reduce the bias of '0' and '1'. B. Sunar quantified the jitter in RO using the URN model and proposed the use of XOR gates to combine signals to improve the jitter region filling rate using XOR gates [14]. However, XOR gate has difficulty in handling the large number of transitions from an oscillating ring. Rozic et al. experimentally investigated this

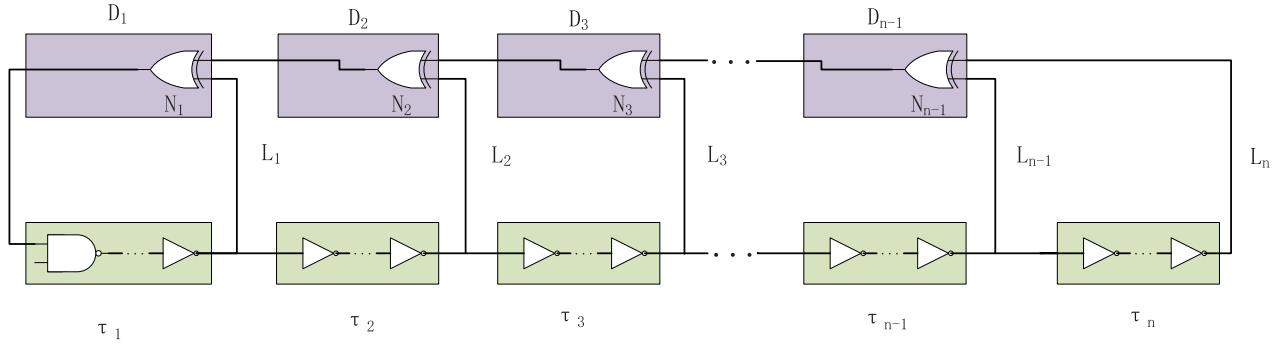


Fig. 2. Simplified delay model for FIRO. τ_i is the delay between the feedback input and the feedback input of the previous stage (τ_1 is the delay between the feedback input of the first stage and the inverter input of the first stage,) L_i is the delay between the feedback logic gate and its feedback input, and D_i is the sum of the component delay of the XOR gate of the i stage and the wiring delay of the input to the next logic gate.

narrow jump suppression effect, showing its deleterious effect on the output [20]. Assuming that the response time of an XOR gate to a single input change is d , the behavior of an XOR gate with a two-input change can be modeled as follows:

$$Z' = \begin{cases} Z, & |t_x - t_y| = 0, \\ Z_{sp}, 0 < |t_x - t_y| < d, \\ \bar{Z}, & |t_x - t_y| \geq d. \end{cases} \quad (1)$$

In Eq. (1), Z is the output voltage of the XOR gate at the time i . Z' is the output behavior of the XOR gate in response to the change in its two inputs. Z_{sp} represents the short pulse output of the XOR gate that has not reached its full voltage due to the high-speed change in A and B. $\bar{Z}$ is the full voltage opposite to the initial state of the XOR. t_x and t_y are the times when the changes occur at the X and Y ports, respectively.

In reality, the jitter occurs in the voltage transitions of logic elements affected by noise; therefore if the two inputs of an XOR gate are pulses arriving through propagation paths of different delays, the behavior model of the XOR gate is also affected by jitter. The behavior model of the XOR gate combined with jitter is as follows:

$$Z' = \begin{cases} Z, & |t_{sxd} - t_{syd} + \Delta t_s| = 0, \\ Z_{sp}, 0 < |t_{sxd} - t_{syd} + \Delta t_s| < d, \\ \bar{Z}, & |t_{sxd} - t_{syd} + \Delta t_s| \geq d. \end{cases} \quad (2)$$

In Eq. (2), t_{sxd} and t_{syd} are the deterministic delays of the delayed paths at the X and Y ends, respectively. Δt_s is the sum of the jitter occurring on both paths, and $\Delta t_s \sim N(0, \sigma_s^2)$.

When $0 < |t_{sxd} - t_{syd} + \Delta t_s| < d$, the XOR gate outputs a short pulse. The amplitude and the phase of the short pulse are related to the jitter in the two input paths. If the jitter decreases $|t_{sxd} - t_{syd} + \Delta t_s|$, the amplitude of the short pulse decreases (as shown by the proximity to the original XOR gate output potential). Conversely, the amplitude of the short pulse increases (manifested as a move away from the original XOR gate output potential). In the input that changes first, the jitter determines the starting phase of the short pulse; in the input that changes later, the jitter determines the ending phase of the short pulse. Therefore, the amplitude and phase of the short pulse contain the jitter information. This means that the

XOR gate converts the jitter into the form of short pulses with random amplitudes and phases.

In existing multi-feedback based TRNGs, XOR gate are embedded in an oscillation ring to increase the oscillation speed of the ring. The multi-feedback structure allows multiple pulses to occur in the ring at the same time, when two pulses reach an XOR gate with an extremely short time difference, the XOR gate will output Z_{sp} . This appears to provide analog randomness to the oscillating ring; however, in the feedback ring currently used, it introduces the risk of periodic oscillation.

B. Time Model of FIRO Pulse Propagation

An FIRO can be obtained by replacing the registers in a Fibonacci configuration linear feedback shift register (LFSR) with inverters. Without register constraints, the oscillation form of a FIRO becomes extremely complex. In order to describe the oscillation form of an FIRO, a simplification model of the delay path the in FIRO is presented in Fig 2.

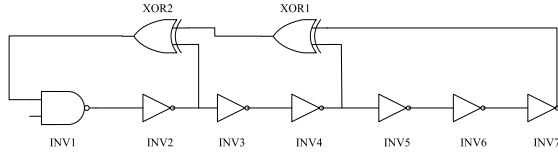
In the FIRO implementation, we use the NAND gate to control the oscillation of the FIRO. When the NAND gate enable input is '0', the NAND gate output is always '1', and oscillation of the FIRO stops. Once the enable input to the NAND gate is '1', the NAND gate works as an inverter in the FIRO and the FIRO starts oscillating. The starting pulse from the NAND gate output is fed to each feedback logic gate through the inverter chain, and eventually the FIRO oscillation is changed through the first feedback logic gate (N_1).

We take the output start pulse of the XOR gate N_1 as the time base. Note T_a as the time node where the start pulse enters N_1 through L_a along the inverter chain. According to the delay model in Fig 2, the time when the pulse first arrives at N_1 along the path L_a can be expressed as follows.

$$T_a = \sum_{i=1}^a \tau_i + L_a + \sum_{j=2}^a D_j \quad (3)$$

The pulse derived from N_1 enters the inverter chain again and the time node that it reach N_1 through L_b is T_{ab} . We represent the time node $T_{abc...}$ for any combination of paths to N_1 by extending the time node subscript.

$$T_{abc...} = T_a + T_b + T_c + \dots + (k-1)D_1 \quad (4)$$


 Fig. 3. FIRO with feedback polynomial of $f(x) = x^7 + x^4 + x^2 + 1$.

If the response time of the XOR gate to any pulse input is 0, the frequency of the FIRO will increase infinitely as the pulse propagates, which is obviously not possible. Suppose that the XOR gate takes a certain time d to respond to the input pulse. When the pulse input to the XOR gate is at a close time point ($T_X - T_Y < d$), the XOR gate produces a narrow pulse suppression effect and outputs short pulses which do not reach its full voltage level. The short pulses with low amplitudes and widths disappear rapidly after propagating into the inverter chain, which eventually manifests itself as a hold in the inverter chain potential. The elimination of short pulses greatly inhibits the effect of the feedback structure in boosting the FIRO oscillation speed and puts the FIRO at the risk of entering periodic oscillation.

By analyzing the time nodes of the pulse input to N_1 , the distance between neighboring time nodes can be observed according to the specific delay parameters of the FIRO. Assuming that short pulses generated by neighboring time nodes at distances less than d_p are eliminated in the inverter chain, we refer to the two time nodes with distances less than d_p and whose distance cannot be made to exceed d_p by jitter in the pulse propagation path as offset pairs. The two time nodes whose distance is close to d_p and whose generated short pulses can pass through the inverter chain affected by jitter are called random pairs. Other time nodes that the generated pulses can definitely pass through the inverter chain are called ordinary time nodes.

Therefore, the form of oscillation of the circuit can be determined by analyzing the type of time nodes in the FIRO. Offset pairs in the FIRO reduce the speed of oscillation, and short pulses containing randomness cannot affect FIRO oscillation through the inverter chain. The presence of random pairs allows the FIRO to enter chaotic oscillations, making the randomness in the jitter easy to sample. Pulses generated by common time nodes can increase the FIRO oscillation speed.

C. FIRO Specific Case Studies

To observe the offset phenomena occurring in an FIRO, we implemented an FIRO with feedback polynomial $f(x) = x^7 + x^4 + x^2 + 1$ on hspice, the specific circuit diagram is shown in Fig 3. The circuit is based on a 45nm process. The XOR gates and inverters in the FIRO are configured to have similar delays so that we can express the delays of each logic gate as 1. Since there is no line delay in hspice, we can estimate the time model of the FIRO in the simplest way. L_i is 0, τ_i and D_i can be expressed as the number of logic gates in it, respectively.

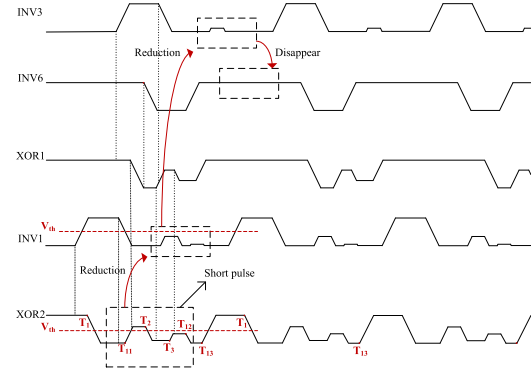


Fig. 4. Critical node waveforms in FIRO.

Thus, the time model of this FIRO can be expressed as

$$\begin{aligned} T_1 &= 2 & T_2 &= 5 & T_3 &= 8 \\ T_{11} &= 5 & T_{12} &= 8 & T_{13} &= 11 \\ T_{21} &= 8 & T_{22} &= 11 & T_{23} &= 14 \\ T_{31} &= 11 & T_{32} &= 14 & T_{33} &= 17 \end{aligned} \quad (5)$$

Obviously, among the time nodes where the starting pulse is extended through the inverter chain, the sub-nodes with the T_1 time node form offset pairs, respectively. The pulses generated at these time nodes cannot propagate further through the inverter chain, so the circuit starts oscillating with a flip of the potential in the inverter chain only at the T_1 and T_{13} time nodes. And at the node T_{13} , only one pulse in the circuit propagates from N_1 to the inverter chain, and the same canceling event occurs at subsequent time nodes of T_{13} as the circuit starts oscillating. This situation where one pulse propagates and spreads in the circuit after a certain time, and the circuit is left with only one pulse again by pulse cancellation causes the FIRO to enter periodic oscillation, with the period being the interval between two pulses appearing at the same position.

The waveform shown in Fig 4 is drawn based on hspice simulation results, it shows the short pulse generated at XOR2 is lost when propagating in the inverter chain. This is because the amplitude of the short pulse at INV1 was reduced to below V_{th} , so the short pulse in INV1 could not be observed in the digital circuit, and the final waveform of the inverter chain shows alternating periodic oscillations of T_1 and T_{13} . At the same time, the randomness contained in the short pulse is completely lost with the elimination of the short pulse.

Since there is no effect of noise in the simulation and random pairs cannot be generated. Assuming that the jitter of a single logic gate follows $(0, \sigma^2)$ in a real circuit, the probability that short pulses from T_2 and T_{11} interval outputs cannot pass through the inverter chain is:

$$\begin{aligned} P(|T_2 - T_{11}| < d_p) &= P(|T_d + \Delta t_s| < d_p) \\ &= \int_{-d_p - T_d}^{d_p - T_d} \frac{1}{\sigma_s \sqrt{2\pi}} e^{-\frac{x^2}{2\sigma_s^2}} dx \end{aligned} \quad (6)$$

From Eq. (5), the delay difference between T_2 and T_{11} is 0 and $T_d = 0$. The paths are separated from the starting pulse

through the first feedback input point, and the number of logic gates of different paths experienced by both T_2 and T_{11} is 3. Therefore, a total of 6 logic gates of jitter are accumulated for both paths. After substituting the specific parameters, it is obtained that:

$$P(|T_2 - T_{11}| < d_p) = \int_{-\infty}^{d_p} \frac{1}{2\sigma\sqrt{3\pi}} e^{-\frac{x^2}{12\sigma^2}} dx \quad (7)$$

In Eq. (7), d_p is much greater than 0. The probability that a short pulse will not pass through the inverter chain is also much greater than 0.5, and T_2 and T_{11} offset periodically during circuit oscillation. The accumulated jitter in T_3 and T_{12} is slightly larger but they still offset with a high probability during oscillation.

From Eq. (6), it can be concluded that the response speed of the XOR gate to input changes, the delay difference between neighboring time nodes, the strength of the jitter in the circuit, and the size of the jitter accumulated at the two time nodes all affect the appearance of the random pairs in an FIRO. The above conclusions can provide good guidance in designing FIRO-based true random number generators.

D. FIRO's Randomness Improvement Principle

In the past, FIRO researchers believed that the introduction of the feedback structure increased the ring oscillation frequency and the rate of jitter accumulation. But our analysis revealed that the ring oscillation frequency did not increase if many offset pairs were present in the ring. Jovan Golic argued that if the change in the feedback signal would not maintain the change in the first inverter input signal, all the inverters in the ring would go to intermediate levels. Nevertheless, short pulses with too small amplitudes and widths are eliminated in the inverter chain and do not help bringing the ring into a meta-stable state.

Consider a random pair affected by clock jitter as a true random source for FIRO. Three approaches are proposed to improve the performance when an FIRO is used as the entropy source in the following ways.

1) *Adjusting the Delay Path:* Control the FIRO to generate random pairs by changing the deterministic delay deviation T_d at its neighboring time nodes so that T_d is close to the response time d_p of the XOR gate to input changes.

2) *Increasing the Number of XORs:* By increasing the number of XOR gates to increase the number of starting pulses diffused through the feedback circuit, more pulses exist in the circuit at the same time to avoid pulse elimination cause periodic oscillations in the circuit and increase the probability of random pair appearance.

3) *Reducing the Response Time of the XOR Gates to Input Changes:* Reducing d_p reduces the generation of offset pairs and increases the oscillation speed of the circuit.

Although an FIRO can be configured so that it enters chaotic oscillations, the suppression of the inverter chain's effect on the oscillation of the XOR gates change circuit reduces the performance of FIRO as an entropy source.

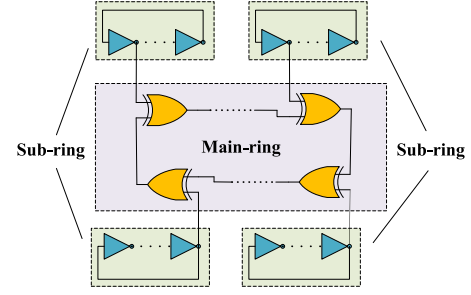


Fig. 5. A novel TRNG architecture.

III. THE PROPOSED TRNG ARCHITECTURE

Due to elimination phenomenon of short pulses in an oscillator, the method of adding XOR gates in a RO does not guarantee increase of jitter accumulation rate in the oscillator. However, short pulses generated by XOR gates provide analog randomness to the circuit, allowing XOR gates to effectively improve the performance of an entropy source. The speed of short pulses generated by an XOR gate is related to the change rate of input potential, and the jitter information carried by short pulses is related to the accumulated jitter of XOR gate input pulses. In order to use short pulses more efficiently to improve the entropy source randomness, we propose a new TRNG entropy source architecture by separating the XOR gate from the inverter and decomposing the single oscillation circuit into sub-rings and main-rings.

Sub-Ring A ring oscillator composed entirely of inverters used to generate a continuous pulse input to the XOR gate of the main-ring.

Main-Ring It consists of all XOR gates that continuously generate short pulses depending on the time difference of the pulses at the input.

The sub-ring and the main-ring collectively form a multi-ring convergent oscillator whose specific structure is defined by the following polynomial.

$$f(x) = \sum_{i=1}^n x_i^{p(i)} \quad (8)$$

In Eq. (8) $n > 1$ and $p(i) > 1$. n denotes the number of XOR gates in the main-ring and $p(i)$ denotes the number of inverter stages in the subring connected by the i -th XOR.

We connect one input of the XOR gate to an independently oscillating sub-ring, increasing the source of jitter generation and improving the speed of short pulse generation by XOR gates. At the same time, the danger of the oscillation circuit stopping oscillation is avoided. The output of each XOR gate is read by a sampling circuit and then heterodyned by an XOR gate to obtain the final random bit output.

A. Short Pulse Utilization Mechanism

In this structure, the short pulse generated by the XOR gate is input to the next level XOR gate. When the potential of the other input of the XOR gate remains unchanged, the XOR gate only behaves as an inverter or a buffer. When a short pulse is input to the XOR, the other input has a potential flip

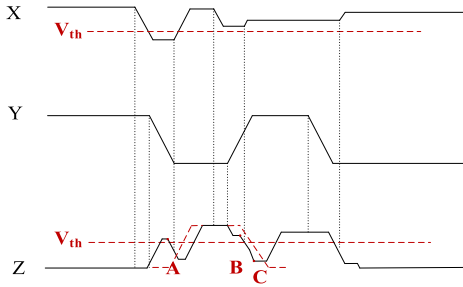


Fig. 6. Effect of short pulses on XOR gate output.

at the same time, and the short pulse will change the effect of the potential flip of the other input on the output of the XOR.

As shown in Fig 6, X and Y are the two inputs of the XOR gate and Z is the output of the XOR, where X is the output of the sub-ring and Y is the output of the previous level of the XOR. In Fig 6, when the Y level changes from '1' to '0', it should originally make Z from '0' to '1' at A, however, since X has a short pulse input, there is a short '01' and '00' for the XOR gate input XY. The '00' causes a delay in the flip of the potential generated by the change in Z for the Y input. Point B shows the effect of the short pulse to advance the potential flip time of Z for the change in Y. Point C shows the faster response of Z to the change in Y input because the level of the input at X does not reach the full voltage level, leaving the output of Z near the middle voltage level.

As seen in Fig 6, the short pulse will change the behavior of the XOR gate in response to a level flip at the other input. Both the amplitude and phase of the output pulse of the XOR gate are affected by the short pulse. By changing the oscillation behavior of the circuit, short pulses will effectively improve the randomness of the oscillator. The output of an XOR gate at the sampling moment is not only related to the phase of sub-rings, but also affected by the short pulses that have not been eliminated in the main-ring for some time before the sampling moment. When a flip-flop samples a short pulse from the output of XOR gate, the short pulse tends to violate the setup and hold time of a flip-flop, causing the flip-flop to become meta-stable. The generation of meta-stability improves the randomness of the entropy source.

B. Design Overview

Based on the proposed TRNG design framework, we use two MRCOs in an FPGA with the polynomial of $f(x) = x_1^1 + x_2^3$ as shown in Fig 5. We use two main-rings, each consisting of two XORs. The two XOR gates are connected to two different sub-rings, a single-stage RO and a three-stage RO. Among them, the single-stage RO can provide the maximum pulse input speed for the XOR gate in the main-ring and the three stages is the minimum number of inverter stages for a RO to output a complete pulse. Since the output of the single-stage RO is a short pulse, it is necessary to place the single-stage RO near the XOR gate connected to it during the layout process to prevent the loss of short pulses by an excessively long path. Similarly, XORs also should be placed adjacent to each other. The single-stage RO consists of

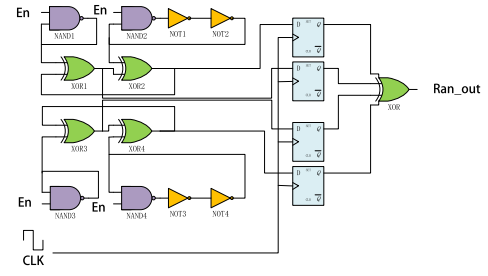


Fig. 7. Specific implementation of TRNG.

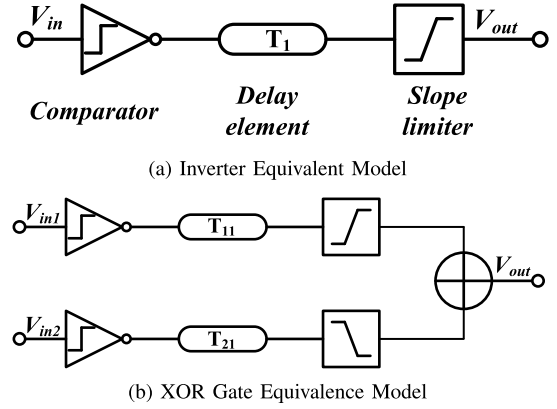


Fig. 8. (a) is the inverter model proposed by Reyner et al. and (b) is the XOR gate model proposed on the basis of (a).

a NAND gate and the three-stage RO consists of a NAND gate and two inverters, with the NAND gate used to turn on and off the oscillation of the TRNG. We use as few logic resources as possible to verify the performance of the proposed design framework, and each logic gate is implemented by the LUT primitive instantiation. The sampling module is a D flip-flop. The clock after PLL multiplication is used as the clock input of the D flip-flop, and the output of each XOR gate is used as the data input of the D flip-flop. The extracted data is directly passed through an XOR gate output to obtain a random bit.

C. Entropy Source Analysis

In order to properly evaluate the quality of the generated random bits, it is necessary to build the corresponding stochastic model to evaluate the performance of entropy sources of the proposed TRNG. Our study uses the physical model of Reyneri et al [23]. to describe the transmission process of short pulses in a logic circuit.

An inverter model is shown in Fig 8(a), it includes an ideal inverting threshold comparator, an ideal non-inertial delayed phase delay T_1 and a slope limiter. The rise and fall time of an inverter is T_2 , and any pulse of duration $P_i < T_2$ will be scaled down in the slope limiter.

As shown in Fig 8(b), a two-input XOR gate equivalent model can be obtained by combining two inverter models, and the slope in opposite directions indicates the logic function of an XOR gate. This model is based on the fact that adjacent pulse inputs always have opposite effects on the output of an XOR gate. To simplify analysis of the relationship between XOR gate potential and input pulses, it is assumed that the

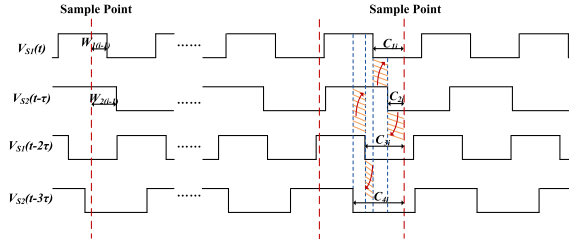


Fig. 9. Timing diagram of an XOR gate input changes during the sampling phase in the proposed TRNG.

variation of XOR gate potential has a constant rate κ . The response of an XOR gate to two adjacent potential changes at inputs can be expressed as follows.

$$V_{xor_1} = V_{xor_0} - \kappa D(T_{in2} - T_{in1} + T_{11} - T_{21}) \quad (9)$$

Eq. (9) holds only if $V_{xor_1} \in [V_{GND}, V_{cc}]$. Where V_{xor_0} is the initial amplitude of the XOR gate when the XOR gate pulse input changes, V_{th} is the threshold voltage of the XOR gate and D is the direction of the XOR gate output change. D can be expressed as follows.

$$D = \frac{V_{in1} - V_{th}}{|V_{in1} - V_{th}|} \frac{V_{in2} - V_{th}}{|V_{in2} - V_{th}|} \quad (10)$$

Considering that the short pulses are scaled down during the propagation of the circuit, the voltage amplitude of the XOR gate at the arrival of the sampled signal is only related to the number of flips of sub-rings and propagation of short pulses for a small period of time. An equivalent model of the response of any XOR gate in main-ring to sub-ring inputs is as follows.

$$V_{xor_1}(t) = \bigoplus_{j=0}^k V_{s_1}(t - 2j\tau) \oplus V_{s_2}(t - (2j + 1)\tau) \quad (11)$$

where $V_{s_i}(t)$ is the amplitude of XOR gate input from sub-ring i to the main-ring at moment t . Since the delays between XOR gates have fixed delay difference, which can be combined with the initial phase difference of sub-rings, we assume in the model that the XOR gates in the main-ring have the same delay τ . j measures the distance between the moment of such voltage change and the current moment, the larger the value of j , the smaller the impact of the short pulse generated by its potential change on the current state of the circuit.

From Eq. (11), the variation of an XOR gate output potential at the arrival of the sampling moment is related to $V_{s_1}(t - 2j\tau)$ and $V_{s_2}(t - (2j + 1)\tau)$. Since short pulses generated by input changes far from the sampling moment are eliminated in the circuit, we only consider the effect of input changes on the XOR gate at $j = 0$ and 1. The timing diagram is shown in Fig 9.

Where w_i is the distance from the sampling point to the nearest edge behind, C_i is the distance from the sampling point to the nearest edge in front. As shown in Fig 9, the voltages at each input port are still changing when the sampling clock arrives, and the different change intervals at the four inputs divide the output change of the XOR gate into four parts.

The transition times of a logic gate obeys a normal distribution with mean μ and variance σ^2 , so k -stage ROs have a half-cycle $X_k \sim N(k\mu, k\sigma^2)$. According to the model proposed by ma et al. [24], let $R_i = \min\{k \mid T_k > s\}$, where $T_k = X_1 + X_2 + \dots + X_k$, meaning R_i is the first increasing k ensuring that T_k is larger than sampling interval s . The probability that $R_i = k$ is as follows.

$$\begin{aligned} Prob(R_i = k + 1) &= Prob(T_k \leq s) - Prob(T_{k+1} \leq s) \\ &= \Phi\left(\frac{(v - k) \cdot \frac{\mu}{\sigma\sqrt{k}}}{1}\right) \\ &\quad - \Phi\left(\frac{(v - k - 1) \cdot \frac{\mu}{\sigma\sqrt{k+1}}}{1}\right) \end{aligned} \quad (12)$$

where $v = s/\mu$ represents the frequency ratio. The distribution of the direction of change D_i of the XOR gate at the sampling moment is

$$\begin{aligned} Prob(D_i = d_i) &= Prob(R_{1i} \bmod 2 = \frac{d_i + 1}{2}) \\ &\quad \cdot Prob(R_{2i} \bmod 2 = \frac{1 - d_i}{2}) \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} Prob(R_{1i} = 2j - \frac{d_i + 1}{2}) \\ &\quad \cdot Prob(R_{2i} = 2k - \frac{1 - d_i}{2}) \end{aligned} \quad (13)$$

For a given R_i , C_i can be expressed as:

$$C_i = s - R_i \quad (14)$$

Combining Eq. (9), Eq. (13) and Eq. (14) we get:

$$\begin{aligned} Prob(B_i = b_i) &= Prob(\Delta t > T_2 | D_i = 2b_i - 1) \\ &\quad + Prob(\Delta t \leq T_2 | D_i = 1 - 2b_i) \\ &\quad (\Delta t = 2C_{i1}^1 - 2C_{i2}^2 + 2C_{i3}^3 - C_{i4}^4) \end{aligned} \quad (15)$$

where 1 in C_i^1 denotes $\min\{C_{1i}, C_{2i}, C_{3i}, C_{4i}\}$, and the value of C_i^m is incremented according to the size of the ordinal number m . Since C_i follows a normal distribution, the introduction of C_{3i} and C_{4i} gives greater variance to Δt . Considering sub-rings change edges at longer times can provide a larger number of C_i to improve the accuracy of the model.

The distribution of the waiting time W_i can be considered to obey a uniform distribution.

$$\begin{aligned} &Prob(b_{i+1}|b_i, \dots, b_1) \\ &= \int_0^1 Prob(b_{i+1}|x) d(Prob(W_i \leq x|b_i, \dots, b_1)) \\ &= K(b_{i+1}) \end{aligned} \quad (16)$$

From Eq. (16), we can calculate the n-bit probability distribution for certain bit patterns.

$$p(b) = Prob(b_n, \dots, b_1) = \prod_{i=1}^n K(b_i) \quad (17)$$

Then the n-bit entropy can be calculated as follows.

$$H_n = \sum_{b \in \{0,1\}^n} -p(b) \log p(b) \quad (18)$$

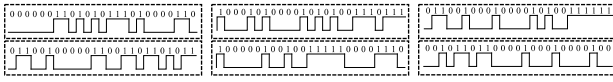


Fig. 10. Results of six restart tests.

In contrast to the conventional TRNG that applies only RO as an entropy source, the XOR gate generates short pulses for the circuit, leading us to consider the effect of short pulses on randomness in the stochastic model. The randomness of sub-rings phase in the proposed TRNG is determined by the quality factor $q = \sigma\sqrt{v}/\mu$ [23], and this randomness can be improved by increasing the sampling interval. The outputs of XOR gates in the main-ring is affected by the time difference of the input phase of sub-rings, producing short pulses with randomness. By simply adding XOR gates to the main-ring with correspondingly connected sub-rings of XOR gates, the sub-rings phase jitter information contained in the short pulses increases with the variance of ΔT , while the speed of generating short pulses can be increased by using a larger number of XOR gates.

IV. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of the proposed TRNG using the NIST SP 800-22 and NIST SP800-90B test suites. To compare the performance of the proposed TRNG in circuits of different processes, the proposed TRNG was implemented in Xilinx Spartan-6 FPGAs and Xilinx Virtex-6 FPGAs, respectively. Due to the lower latency caused by the more advanced process level, two different sampling clocks of 350 MHz and 500 MHz were set for Spartan-6 and Virtex-6.

A. Restart Test

To verify whether the output sequence of the proposed TRNG is a pseudo-random number, we collected the output sequence after six restarts, as shown in Fig 10. The experimental results show that the data sampled after six restarts are different, which proves that the random sequence generated by the proposed TRNG has true randomness.

B. Autocorrelation Test

The degree of correlation between bits generated at different times in a random sequence can be measured by the autocorrelation coefficient. We collected one million consecutive bits in Spartan-6 and Virtex-6, respectively, and ran the autocorrelation function in Python. The lag values in the autocorrelation function were set to 1 to 100, and the autocorrelation factor (ACF) between variables with lags of 1 to 100 bits was analyzed. The test results are shown in Fig 11, where the ACF fluctuates between 0 and 0.004. The results indicate that the proposed TRNG indeed generates random sequences with very low autocorrelation and can effectively resist the correlation analysis attack.

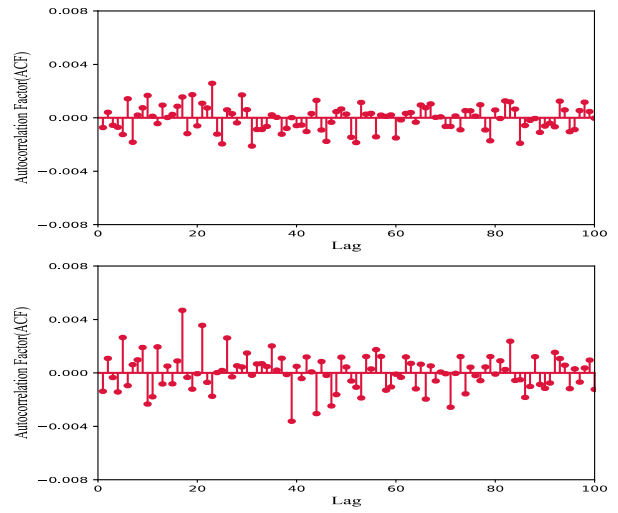


Fig. 11. Autocorrelation test for one million bits of data.

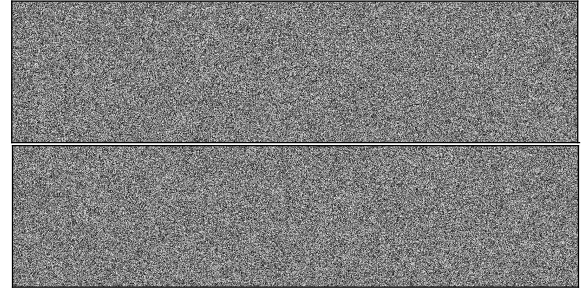


Fig. 12. Bias test for one million bits of data.

C. Bias Test

A qualified random number sequence should have desirable statistical properties, and the distribution of ‘0’ and ‘1’ in one million bits of data is shown in Fig 10. Here white pixel dots denote ‘1’ and black pixel dots denote ‘0’, the black and white pixel dots shown in the figure are evenly distributed, reflecting the good statistical properties of the generated data. The statistical deviation can be calculated by the following equation.

$$Bias = \frac{|N_0 - N_1|}{2 \times 1000000} \quad (19)$$

In Eq. (16) N_0 and N_1 denote the numbers of ‘0’ and ‘1’ in one hundred consecutive bits, respectively. From Eq. (19), the statistical deviation was calculated to be 0.0454% in Spartan-6 and 0.0266% in Virtex-6. This proves that the proposed TRNG is capable of generating random number sequences with low bias.

D. Nist Test

NIST SP 800-22 is a relatively comprehensive and rigorous test suites available for random number testing [21]. It contains 15 tests for evaluating TRNG performance, and the tests calculate a p-value for the random sequence. If the p-value is below 0.01, it indicates that the random sequence is poorly

TABLE I
TEST RESULTS FOR NIST SP 800-22 ON TWO DEVELOPMENT BOARDS

Test	Spartan6		virtex6	
	P-value	Prop	P-value	Prop
ApproximateEntropy	0.199045	0.985	0.448424	0.986
BlockFrequency	0.073872	0.995	0.138069	0.990
CumulativeSums	0.05166	0.982	0.445449	0.987
FFT	0.342451	0.995	0.645448	0.981
Frequency	0.190654	0.991	0.823725	0.985
LinearComplexity	0.846339	0.993	0.428095	0.989
LongestRun	0.911413	0.994	0.360287	0.992
NonOverlappingTemplate	0.493228	0.990	0.498768	0.990
OverlappingTemplate	0.009136	0.980	0.622546	0.987
RandomExcursions	0.506098	0.988	0.502287	0.990
RandomExcursionsVariant	0.648175	0.988	0.498154	0.985
Rank	0.899171	0.994	0.992670	0.990
Runs	0.048404	0.989	0.406499	0.992
Serial	0.0580325	0.988	0.180025	0.989
Universal	0.979788	0.983	0.083018	0.986

* For tests with two or more subtests, the p-value is the average of all subtests

TABLE II
TEST RESULTS FOR NIST SP 800-90B ON TWO DEVELOPMENT BOARDS

Test	Spartan6		Virtex6	
	p-max	h-min	p-max	h-min
MCV	0.500606	0.998251	0.50046	0.998675
Collision	0.51931	0.944718	0.519531	0.944718
Markov	3.01702E-39	0.999704	3.12732e-39	0.999299
Compression	0.5	1	0.5	1
t-Tuple	0.522488	0.93653	0.521444	0.939416
LRS	0.502832	0.991852	0.500161	0.999536
Multi-MCW	0.500207	0.999403	0.500264	0.999237
Lag	0.501607	0.99537	0.500486	0.998598
Multi-MMC	0.500575	0.998342	0.500637	0.998162
LZ78Y	0.500531	0.998469	0.524108	0.932064

randomized. We collected 1000 sets of data of one million bits in length in different development boards to perform the tests, and the results are shown in Table I. Sequences pass random tests with a very high pass rate, indicating that the proposed TRNG has good randomness.

The non-independent identical distribution test of the NIST SP800-90B test evaluates the randomness of the data by ten minimum entropy estimators and finally uses the smallest value among all estimators as the minimum entropy [22]. The size of the minimum entropy measures the randomness of a random sequence, and the experimental results show that the minimum entropy of our proposed TRNG is 0.93653 and 0.932064, respectively. The complete test results are shown in Table II.

E. Frequency Injection Test

Ugo Mureddu demonstrated the phenomenon that the output random numbers of an oscillating ring-based TRNG fail the statistical test when the oscillating ring goes into

TABLE III
NIST SP 800-90B TEST RESULTS UNDER TWO
FREQUENCY INJECTION ATTACKS

Test	1-RO Inject		3-ROs Inject	
	p-max	h-min	p-max	h-min
MCV	0.500476	0.998628	0.500526	0.998483
Collision	0.51953	0.944718	0.511719	0.966577
Markov	3.40721E-39	0.998333	3.17227e-39	0.999138
Compression	0.507812	0.977632	0.5	1
t-Tuple	0.520930	0.940840	0.520727	0.941401
LRS	0.501625	0.995320	0.518212	0.948386
Multi-MCW	0.500278	0.999197	0.500457	0.998681
Lag	0.500583	0.998320	0.500264	0.999238
Multi-MMC	0.500304	0.999124	0.500367	0.998942
LZ78Y	0.500547	0.998422	0.500281	0.999191

lock [25]. We implemented two frequency injection attacks on the proposed TRNG by placing 3000 single-stage RO and 3000 three-stage RO near MRCO to observe whether the randomness of the TRNG output is affected when the frequency of the sub-rings is locked. The data sampled under both attack scenarios were tested with NIST SP800-90B and the results are shown in Table III. The entropy of the random sequence is not reduced by the frequency injection attack on the sub-rings, and the oscillatory behavior of the main-ring is non-periodic, so the proposed TRNG is resistant to the frequency injection attack.

F. Temperature and Voltage Test

The stability of the proposed TRNG to temperature and voltage variations was tested by changing the voltage and temperature in Virtex-6 and monitoring them with the chip-scope. The range of the tested voltage and temperature was from 20°C to 80°C and 0.9V to 1.2V, respectively. One million consecutive bits were collected for the TRNGs in different environments and the statistical bias was calculated, the results show that the structure with polynomial $f(x) = x_{xor1}^1 + x_{xor2}^3$ appears to have a large statistical bias variation for temperature and voltage variations. At this time, the output random sequence does not pass the NIST SP 800-22 test.

We added the number of connected sub-rings in the main-ring and tested the MRCO with polynomial $f(x) = x_1^1 + x_2^3 + x_3^1 + x_4^3$. The results show that increasing the number of sub-rings can effectively improve the robustness of the proposed TRNG to temperature and voltage variations.

G. Performance Comparison

Table IV compares the performance of the proposed TRNG with state-of-the-art TRNGs implemented in FPGAs. The proposed TRNG shows the highest throughput rate among all of the recent TRNGs on implementation in FPGAs. We report the achievable throughput rates per slice for these TRNG designs, which are used to evaluate the performance of the entropy source. The results show that even in the poorly processed Spartan6, MRCO still exhibits much higher performance than

TABLE IV
IMPLEMENTATION RESULTS AND COMPARISON WITH OTHER STATE-OF-THE-ART FPGA TRNG DESIGNS

Entropy Source	FPGA Family	Resource Cost	Throughput [Mbps]	Throughput/Slice [Mbps/Slice]	Power [mW]	Post-processing
LRO [4]	Spartan-6	4LUTs/3FFs	0.76	0.76	-	No
DCFL [7]	Cyclone-IV	298LUTs	150	2.013	-	Yes
MSFRO [8]	Virtex-6	24LUTs/2FFs	290	48.33	3703	No
JL-STR [9]	Virtex-6	56LUTs/19FFs	100	7.143	1.15	Yes
TROT [10]	Artix-7	32LUTs/55FFs/17CARRT4	25	3.125	8.843	Yes
PDL-RO [11]	Spartan-3	528LUTs/177FFs	6	0.045	-	Yes
GARO [12]	Artix-7	50LUTs/79FFs	280	22.4	-	Yes
RO [15]	Cyclone II	83LUTs/26FFs	100	4.819	-	Yes
MFRO [18]	Spartan-6	-	125	-	-	No
COR [19]	Artix-7	25LUTs/37FFs	160	17.297	-	No
This work	Spartan-6	13LUTs/4FFs	350	107.692	110	No
	Virtex-6	13LUTs/4FFs	500	153.846	112	No

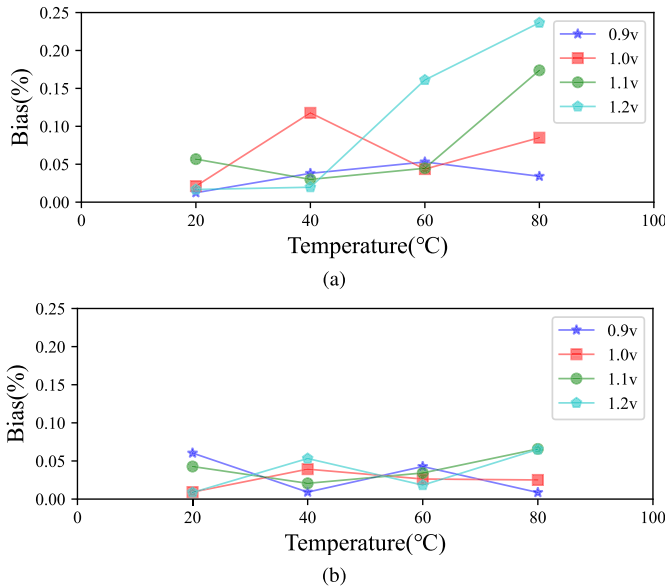


Fig. 13. Results of temperature and voltage tests.

other entropy source structures. As shown in Table IV, the meta-stable-based entropy source structures - LRO [4] and DCFL [7] can produce throughput rates of only 0.76 and 2.013 per slice, which reflects that entropy source structures that rely on meta-stability as the source of randomness need to consume a large amount of hardware resources to obtain adequate throughput rates. By utilizing jitter in conventional ROs as the source of randomness, the throughput rate that can be generated per slice can reach 4.819 [15]. The performance of the entropy source is significantly improved when XOR gates are embedded in ROs [12], [18], [19]. JL-STR and TROT adopt different entropy source structures from ROs, and their way of jitter accumulation is better than RO, but its throughput rate is still lower than that of ROs containing the XOR gates [9], [10]. Recently, MSFRO has effectively increased the accumulation rate of jitter in the oscillator by adding multi-stage feedback in a RO, which can produce a throughput rate

of 48.33 per slice. Compared to the MSFRO, our proposed MRCO in Virtex-6 achieves a throughput rate of 153.846 per slice, which is three times that of the MSFRO. The power consumption of the MRCO was evaluated with the Xilinx XPower Analyzer, and due to the use of the PLL as the sampling clock, the PLL has 95mW of power consumption and the logic part consumes only 0.28mW. Overall, our proposed entropy source architecture provides the highest throughput rate with the least hardware resources compared to other related architectures.

V. CONCLUSION

In this paper, a TRNG is designed based on the new MRCO entropy source structure. we have investigated the mechanism of the XOR gate to enhance the randomness in an oscillatory ring. The propagation of pulses in an FIRO and the cancellation phenomenon of short pulses generated by the XOR gate in the inverter chain are analyzed by a simple time model. The proposed MRCO separates the XOR gates and inverters in an oscillator, in order to efficiently generates short pulses for randomness enhancement. Experimental results in FPGAs have shown that the proposed entropy source structure effectively has a very high performance. The throughput rate can reach 500 Mbps on Virtex-6 FPGA using only 13 LUTs and 4 FFs, and it passes all the statistical tests of NIST without post-processing. Comparing the performance of the proposed TRNG with state-of-the-art TRNG designs implemented in FPGAs, the proposed TRNG design achieve higher throughput rates with less hardware overhead. The proposed TRNG is insensitive to frequency injection attacks, and it can combat temperature and voltage variations by increasing the structural complexity.

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